



Solid–electrolyte interphase formation during Li metal deposition in $\text{LiN}(\text{SO}_2\text{F})_2$ -based solvate ionic liquids

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Received: 6 January 2024 / Revised: 14 February 2024 / Accepted: 15 February 2024 / Published online: 21 February 2024
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Abstract

Solvate ionic liquids (SILs) are promising electrolytes for Li metal batteries. In this study, Li plating-stripping reactions in SILs were investigated using electrochemical quartz crystal microbalance measurements and X-ray photoelectron spectroscopy (XPS). SILs were prepared by mixing Li salt and glyme (triglyme or tetraglyme) in a 1:1 molar ratio. During the Li plating-stripping reaction, a reversible mass change was observed in $\text{LiN}(\text{SO}_2\text{F})_2$ (LiFSA) based SILs. In contrast, a considerably higher mass change than the theoretical value calculated from the Faraday current was observed in the $\text{LiN}(\text{SO}_2\text{CF}_3)_2$ (LiTFSA) based SILs, owing to the accumulation of decomposition products of electrolytes on the electrode surface. XPS depth profiling for the deposited Li suggested the formation of thin solid-electrolyte interphase (SEI) in the LiFSA-based SILs while considerably thicker SEI was formed in the LiTFSA-based SILs. The SEI formed in LiFSA-based SILs effectively suppressed the decomposition of electrolytes. Thus, LiFSA-based SILs are favorable for achieving highly reversible charge–discharge of Li electrodes in Li metal batteries.

Keywords Solvate ionic liquids · Concentrated electrolytes · EQCM · Li metal batteries

Introduction

Nonaqueous electrolyte solutions derived from the dissolution of Li salts in organic solvents exhibit relatively high ionic conductivities and oxidation/reduction stabilities and are widely used in Li-ion batteries (LIBs) [1, 2]. The ionic conductivity of an electrolyte solution is determined as the product of the number of ion carriers and their mobility [3]. Increasing the concentration of Li salts increases the number of carriers but decreases ion mobility owing to the increase in viscosity [4]. This trade-off relationship results in the maximum ionic conductivity of the electrolyte solution at a salt concentration of approximately 1 mol dm^{-3} , which is commonly used in LIBs [5]. However, highly concentrated

electrolytes (HCEs) with Li salt concentrations exceeding 3 mol dm^{-3} have recently gained attention owing to their unique properties [6–8]. Almost all solvents participate in the coordination of Li ions in HCEs, resulting in the low activity of free (uncoordinated) solvents [9–11]. This low free-solvent activity of HCEs results in improved oxidation stability, [12] reduced flammability, [13] reduced solubility of the intermediates of electrochemical reactions, [14, 15] and unique ion conduction properties [6, 16].

Among HCEs, equimolar mixtures of glymes such as triglyme (G3), tetraglyme (G4), and Li salts are similar to ionic liquids. The chelate effect causes Li^+ and glyme to form a stable solvate cation ($[\text{Li}(\text{glyme})]^+$) in these equimolar mixtures [13]. Because the lifetime of the complex cation is relatively long, these mixtures can be regarded as ionic liquids composed of $[\text{Li}(\text{glyme})]^+$ and counter anions, and are referred to as “solvate ionic liquids” (SILs) [17–19]. In HCEs, solvents and anions competitively interact with Li^+ ; thus, it is important to use anions with low Lewis basicity to maintain the stable complex structure ($[\text{Li}(\text{solv})_n]^+$) [20]. Because bis(trifluoromethanesulfonyl)amide (TFSA) anion, $\text{N}(\text{SO}_2\text{CF}_3)_2^-$, exhibits low Lewis basicity, good electrochemical stability, and high solubility in various organic solvents, it is widely used as a salt in HCEs [7, 8,

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21]. Bis(fluorosulfonyl)amide (FSA) anion, $\text{N}(\text{SO}_2\text{F})_2^-$, is now frequently used to prepare HCEs [22]. Because the S–F bond in FSA^- is more easily cleaved than the S– CF_3 bond in TFSA^- , FSA^- decomposes at the electrode/electrolyte interface [23]. The aggregation of Li^+ and FSA^- ions, in particular, lowers the energy level of the lowest unoccupied molecular orbital (LUMO) of FSA^- , resulting in the selective decomposition of FSA^- to form a passivation film (solid–electrolyte interphase; SEI) on the negative electrode surface [24]. The FSA-derived SEI is effective in suppressing subsequent electrolyte decomposition, and highly reversible Li deposition and dissolution are possible at the negative electrode in Li metal batteries using LiFSA-based HCEs [25]. Most of the reports on FSA-derived SEI formed on Li metal, the SEI has been characterized ex-situ, that is, after disassembling the electrochemical cells, and studies on in-situ analysis are rare [26]. In this study, we used an electrochemical quartz crystal microbalance (EQCM) to investigate the SEI formation process during Li metal plating–stripping in LiFSA-based SILs and compared it with that in LiTFSA-based SILs.

Experimental

[Li(glyme)]TFSA and [Li(glyme)]FSA (Fig. 1) were prepared by mixing G3 or G4 (Nippon Nyukazai Co., Ltd.) with LiTFSA (Solvay, Japan) or LiFSA (Kishida Chemical) in a molar ratio of 1:1 in an Ar-filled glovebox (VAC , $[\text{H}_2\text{O}] < 1$ ppm). An HZ-7000 potentiostat (Hokuto Denko Corp.) was used in conjunction with a QCM system (QCM922A, Seiko EG&G) to perform EQCM measurements. Air-tight two-electrode EQCM cells (VQ3A, EC Frontier) were assembled in an Ar-filled glovebox. An AT-cut 9 MHz quartz crystal coated with Ni or Cu (5 mm in diameter, mirror-polished;

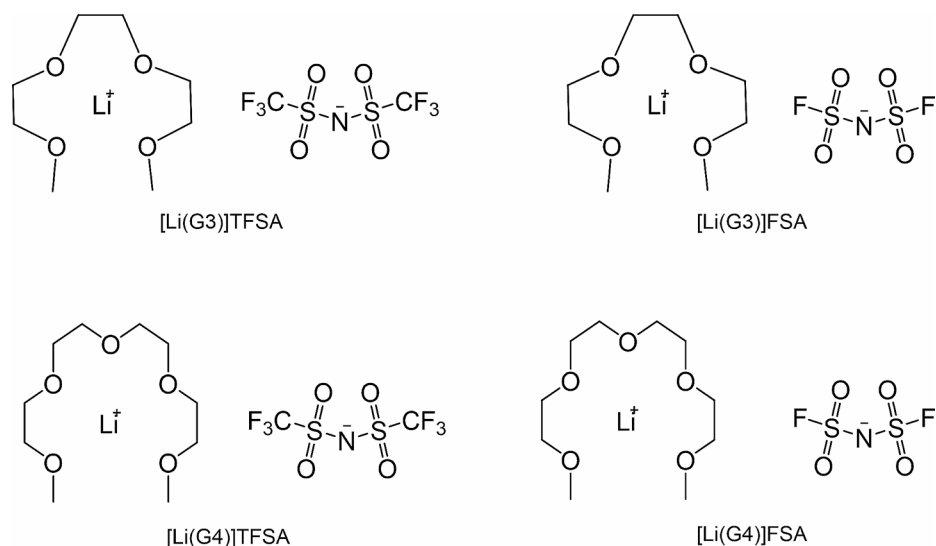
Seiko EG&G) and Li foil (Honjo metal, 200- μm thick) were used as the working and counter electrodes, respectively. Cyclic voltammetry (CV) coupled with QCM was conducted in an environmental chamber (SU-242) at 30 °C. The following Sauerbrey equation was used to calculate the mass change from the frequency change in the quartz crystal:

$$\Delta f = \frac{-2f_0^2}{A\sqrt{\mu_q\rho_q}}\Delta m$$

where Δf , f_0 , A , μ_q , ρ_q , and Δm represent the frequency change, fundamental resonance frequency (9.00×10^6 Hz), surface area of the electrode, shear modulus of quartz (2.95×10^{10} kg m^{-1} s^{-2}), density of quartz (2.65×10^3 kg m^{-3}), and mass change, respectively. The resonance resistance (R) was measured during the CV measurements. Although it is known that R and Δf are correlated, [26] R -correction to Δf was not performed in this study for simplicity, because two competing effects govern the resonance resistance changes; the reduced viscosity and density of the electrolytes near electrode surfaces decrease R during Li deposition, whereas the higher roughness of the deposited Li increases R .

X-ray photoelectron spectroscopy (XPS) was performed on the Cu foil after electrochemical Li deposition for 5 h at 0.1 mA cm^{-2} in a 2032 coin-type cell using a PHI Quantera SXM Ulvac-PHI spectrometer. Following the galvanostatic deposition, the coin cell was disassembled in the glovebox, and the electrode was washed with anhydrous tetrahydrofuran (Wako Pure Chemical Industries, Ltd.) to remove residual electrolyte. Subsequently, the electrode was placed in an airtight transfer chamber and subjected to XPS analysis without exposure to air. XPS were carried out with an Al

Fig. 1 Chemical structures of SILs used in this study



K α X-ray source. To conduct depth profiling, Ar-ion etching was used, and the etching rate was 23 nm min⁻¹ based on SiO₂ surface. The binding energy was not calibrated because adventitious carbon was not observed after Ar ion etching; however, a charge neutralizer was used throughout the experiments.

Results & discussion

Table 1 summarizes the physicochemical properties of the previously reported SILs [20, 27]. The Li⁺ concentration in [Li(glyme)]FSA is higher than that in [Li(glyme)]TFSA because FSA has a lower molecular weight and smaller anion size compared with TFSA. The viscosity of [Li(glyme)]FSA is slightly lower than that of [Li(glyme)]TFSA owing to the smaller hydrodynamic radius of the FSA⁻ anion. Owing to its higher molar concentration and lower viscosity, [Li(glyme)]FSA exhibits a slightly higher ionic conductivity than [Li(glyme)]TFSA. Compared with G3-based SILs, G4-based SILs exhibit lower viscosities. Glymes are flexible molecules, and the number of conformers in [Li(G4)]⁺ is greater than that in [Li(G3)]⁺ [28]. Furthermore, the Coulombic interaction between the ions in the G4-based SILs is weaker than that in the G3-based SILs because of the lower concentration of Li salt in the G4-based SILs. These factors result in the lower viscosity of G4-based SILs, resulting in higher ionic conductivity compared with G3-based SILs. The self-diffusion coefficient ratio between Li⁺ and glyme ($D_{\text{glyme}}/D_{\text{Li}}$) obtained from pulsed-field gradient nuclear magnetic resonance (PFG-NMR) spectroscopy is almost unity for [Li(glyme)]TFSA and [Li(glyme)]FSA, indicating that Li⁺ and glyme diffuse together in solution [20]. This observation implies that the solvate cation ([Li(glyme)]⁺) is stable and long-lived complex cation with a significant lifetime, and that there is almost no free solvent in these SILs. Furthermore, a certain fraction of anions form contact ion pairs (CIPs) with Li⁺ ions in both G3- and G4-based SILs, according to previously reported Raman spectra [27, 29]. Therefore, the liquid structures of these SILs are extremely similar.

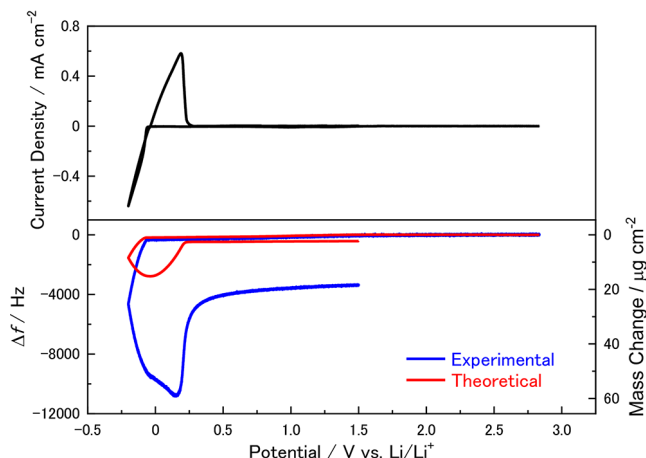
Table 1 Molar concentration (c), viscosity (η), ionic conductivity (σ), and diffusion coefficient ratio ($D_{\text{glyme}}/D_{\text{Li}}$) in [Li(G3)]TFSA, [Li(G3)]FSA, [Li(G4)]TFSA and [Li(G4)]FSA at 30 °C. Data are obtained from previous publications [20, 27]

	$c / \text{mol dm}^{-3}$	$\eta / \text{mPa s}$	$\sigma / \text{mS cm}^{-1}$	$D_{\text{glyme}}/D_{\text{Li}}$
[Li(G3)] TFSA	3.06	169	1.10	0.94
[Li(G3)]FSA	3.71	139	1.65	1.00
[Li(G4)] TFSA	2.75	106	1.60	0.98
[Li(G4)]FSA	3.25	94	1.98	1.01

Figure 2 shows the initial cycle of CV and the resonance frequency change (Δf) obtained during the Li metal plating-stripping process on Ni-coated quartz crystal electrodes in [Li(G4)]TFSA and [Li(G4)]FSA. The mass change calculated from Δf using the Sauerbrey equation is also shown on the vertical axis. The theoretical mass change calculated from the Faraday current is also shown in Fig. 2. This calculation assumes that the Faraday current is solely due to Li deposition/stripping and that no side reactions occur. A weak reduction current is observed at ~1.5 V during the cathodic scan of electrode potential. This is due to the reductive decomposition of the electrolyte components, and the increase in mass indicates that the decomposition products are deposited on the electrode surface to form a surface film. The amount of the deposited films are approximately 0.9 and 2.3 $\mu\text{g cm}^{-2}$ for [Li(G4)]TFSA and [Li(G4)]FSA, respectively, at 0.5 V vs. Li/Li⁺, indicating that more reductive decomposition products are deposited in [Li(G4)]FSA. Similar trends are observed for the G3-based SILs ([Li(G3)]TFSA and [Li(G3)]FSA), as shown in Fig. S1. Ethers (including glymes) are known to have good reductive stability, [1, 13] therefore, the mass changes observed prior to Li deposition can be attributed to the reductive decomposition of anions. The LUMO energy levels of the TFSA⁻ and FSA⁻ anions decrease in concentrated electrolytes owing to CIP formation with Li⁺, which can induce reductive decomposition of the anions on the Ni electrode [24, 30]. The greater mass change in the FSA-based SIL at > 0.5 V vs. Li/Li⁺ during the cathodic scan implies that compared with TFSA, FSA exhibits inferior reductive stability. During the cathodic scan, a reduction current associated with Li metal deposition is observed below 0 V, and an oxidation current is observed in the reverse scan, corresponding to the stripping of the deposited Li. Both electrolytes exhibit a current density of approximately -0.6 mA cm^{-2} at -0.2 V , which can be attributed to their similar Li⁺ concentration, ionic conductivity and viscosity. The similarity in these parameters results in comparable charge-transfer kinetics and mass transport dynamics. During the initial CV cycle, the Coulombic efficiencies (anodic total charges divided by cathodic total charges) for [Li(G4)]TFSA and [Li(G4)]FSA are 84.9 and 68.9%, respectively, indicating that the reductive stability of FSA is inferior to that of TFSA. The LiFSA/sulfolane-based concentrated electrolyte exhibits a higher Coulombic efficiency; [31] therefore, further increasing the LiFSA concentration in the glyme-based electrolyte, or continuing with cell cycling may also increase electrochemical reversibility. The lower Coulombic efficiency observed for [Li(G4)]FSA may be due to the incomplete coverage of SEI formed on the electrode surface during the initial cycle.

The weight gain in [Li(G4)]TFSA calculated from the change in resonance frequency during Li plating-stripping

(a) [Li(G4)]TFSA



(b) [Li(G4)]FSA

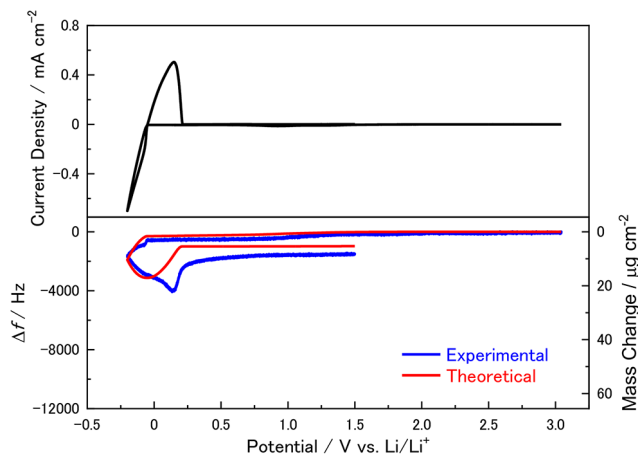
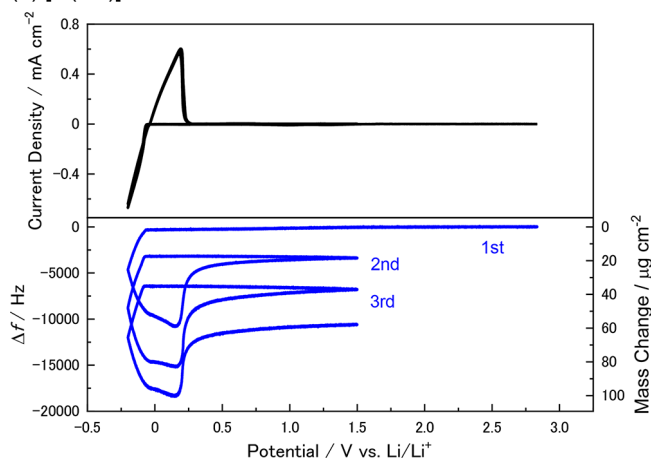


Fig. 2 Initial CVs (black line) and resonance frequency change (Δf) (blue line) recorded using a QCM with a Ni-coated quartz crystal electrode in (a) [Li(G4)]TFSA and (b) [Li(G4)]FSA measured at a scan rate at 0.5 mV s^{-1} at 30°C . The mass change calculated from Δf using

the Sauerbrey equation is also shown on the vertical axis. The theoretical mass change (red line) calculated from the current of CVs is also shown

(a) [Li(G4)]TFSA



(b) [Li(G4)]FSA

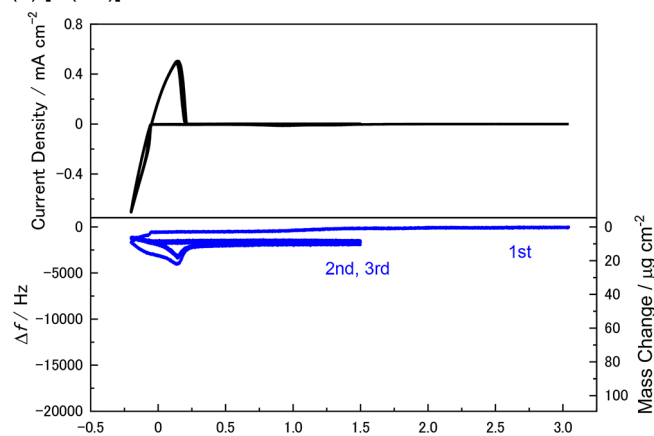


Fig. 3 Cycle dependence of resonance frequency change (Δf) during CV scans with a Ni-coated quartz crystal electrode in (a) [Li(G4)]TFSA and (b) [Li(G4)]FSA measured at a scan rate of 0.5 mV s^{-1} at

30°C . The mass change calculated from Δf using the Sauerbrey equation is also shown on the vertical axis

is more than three times greater than that predicted from the Faraday current observed during CV (Fig. 2), which is consistent with previously reported results for the [Li(G3)]TFSA electrolyte [26]. This is because an effective SEI is not formed in [Li(G4)]TFSA, and the Li surface continuously reacts with the electrolyte, resulting in the accumulation of decomposition products on the electrode surface. The weight gain continues even during the Li stripping process ($E > 0 \text{ V}$), implying that electrolyte decomposition continues via a chemical reaction between the Li surface and the electrolyte. Furthermore, the weight loss during the anodic scan ($\sim 40 \mu\text{g cm}^{-2}$) is greater than the weight gained from the deposited Li ($\sim 12 \mu\text{g cm}^{-2}$; calculated based on the Faraday current), suggesting that the SEI on the electrode

surface is either exfoliated or dissolved during the Li stripping reaction. Although the weight gain in the [Li(G4)]FSA electrolyte is slightly higher than the theoretical value calculated from the CV current, it is considerably lower than that in the [Li(G4)]TFSA electrolyte. This suggests that the SEI derived from FSA[−] on the Li surface effectively suppresses continuous electrolyte decomposition [23, 32–35].

Figure 3 depicts the CV curves and weight changes during cycling. In [Li(G4)]TFSA, a continuous mass increase of approximately $19 \mu\text{g cm}^{-2}$ per cycle is observed. In contrast, an increase of $9 \mu\text{g cm}^{-2}$ is observed only during the first cycle for [Li(G4)]FSA, and no increase is observed after the second cycle. This observation also indicates that the passivation layer forms via the decomposition of

the FSA^- anion, which inhibits the accumulation of electrolyte decomposition products on the electrode surface. Figure S2 depicts the change in resonance resistance during the plating-stripping process. The resonance resistance in both electrolytes increase with Li deposition, followed by a decrease during the dissolution reaction. It is known that the resonance resistance increases with an increase in the local viscosity and density near the electrode surface, as well as an increase in electrode roughness [26]. The concentration of Li salt near the electrode surface decreases during the Li deposition reaction, resulting in a decrease in viscosity and density. The resonance resistance decreases as the viscosity and density decrease; however, the resonance resistance increases during Li deposition (Fig. S2). These findings indicate that the deposited Li and/or SEI formed on the electrode surface has relatively rough surfaces, which is similar to previously reported results [25].

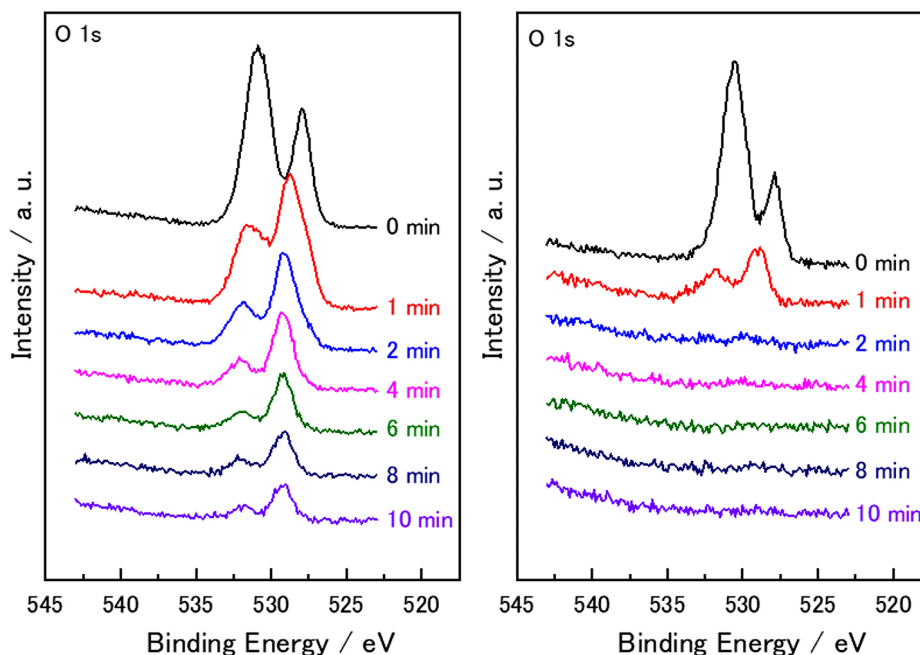
Figure 4 shows the O1s XPS spectra of the Li-deposited Cu foil in the $[\text{Li}(\text{G3})]\text{TFSA}$ and $[\text{Li}(\text{G3})]\text{FSA}$ electrolytes. Figure S3 depicts the XPS spectra of C1s, F1s, Li1s, S2p, and N1s. Ar-ion etching was used to evaluate the thickness of the SEI layer on the deposited Li. The peaks at 531 and 528 eV are attributed to ROLi and C-O/O-C=O species, respectively, [36, 37] which are formed on the deposited Li via the reductive decomposition of the electrolytes. Here we do not focus on the detailed assignment of the spectra because: (i) binding energy calibration could not be performed for depth profiling (due to the removal of adventitious carbon after 1 min etching), (ii) depth profiling provides limited signal-to-noise ratio except for O1s spectra as shown in Fig. S3. Ar-ion etching is known to easily damage the SEI formed on the electrodes, with the extent of

damage varying across different elements [38]. The intensities of the XPS signals observed for samples deposited in $[\text{Li}(\text{G3})]\text{TFSA}$ and $[\text{Li}(\text{G3})]\text{FSA}$ decrease through Ar-ion etching, indicating that the SEI is removed. For the sample deposited in the $[\text{Li}(\text{G3})]\text{TFSA}$ electrolyte, the peaks remain visible even after 10 min etching, whereas the peaks observed for the sample deposited in $[\text{Li}(\text{G3})]\text{FSA}$ completely disappear after 2 min etching. This result suggests that the SEI formed in the $[\text{Li}(\text{G3})]\text{FSA}$ electrolyte is thinner than that formed in the $[\text{Li}(\text{G3})]\text{TFSA}$ electrolyte, which is consistent with the EQCM results.

Conclusion

Using EQCM and XPS measurements, we investigated the SEI formation during Li deposition/stripping in SILs ($[\text{Li}(\text{glyme})]\text{TFSA}$ and $[\text{Li}(\text{glyme})]\text{FSA}$). During Li deposition, a three-fold higher increase in mass was observed in $[\text{Li}(\text{glyme})]\text{TFSA}$ compared with that predicted by the current observed during CV. This mass gain persisted during cycling, suggesting continuous decomposition of the electrolyte on the deposited Li surface and the accumulation of its decomposition products on the electrode surface. In contrast, in $[\text{Li}(\text{glyme})]\text{FSA}$, the mass change observed by QCM was close to that predicted by the CV current. Furthermore, a reversible mass change was observed during cycling, indicating a stable Li plating-stripping reaction owing to the effective SEI formed during the initial deposition of Li by the reductive decomposition of FSA^- . The depth profile of the XPS spectra also confirmed the formation of a thinner SEI in $[\text{Li}(\text{glyme})]\text{FSA}$ than that in

Fig. 4 XPS spectra of the O1s photo-emission lines collected from Cu foil after Li-deposition at 0.1 mA cm^{-2} for 5 h in $[\text{Li}(\text{G3})]\text{TFSA}$ and $[\text{Li}(\text{G3})]\text{FSA}$ electrolytes. Ar-ion etching was performed to conduct depth profiling, under the etching rate of 23 nm min^{-1} based on SiO_2 surface. The binding energy was not calibrated because adventitious carbon was not observed after Ar ion etching, however, a charge neutralizer was utilized throughout the experiments



[Li(glyme)]TFSA. Therefore, FSA-based SILs are favorable for achieving highly reversible charge–discharge of Li electrodes in Li metal batteries.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10008-024-05843-4>.

Acknowledgements This study was funded in part by JSPS KAKENHI (Grant No. JP19H05813, JP21H04697, JP22H00340, and JP23K17370) from the Japan Society for the Promotion of Science (JSPS) and JST ALCA-SPRING (Grant Number JPMJAL1301), Japan. The authors would like to thank Prof. Nobuyuki Serizawa and Prof. Yasushi Katayama of Keio University for their insightful advice on establishing the EQCM setup.

Funding Open Access funding provided by Yokohama National University.

Declarations

Competing interest The authors declare no competing financial interest.

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